INCH-POUND

MIL-T-62414(AT)
23 June 1989
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

TEST EQUIPMENT (SIMPLIFIED)
SET COMMUNICATOR (SET COM)

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the set communicator (Set Com) for the Simplified Test Equipment-Expandable (STE-X) test set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, Mi 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-E-6051	- Electromagnetic Compatability Requirements, Systems.
MIL-P-11268	- Parts, Materials and Processes Used in Electronic Equipment.
MIL-T-28800	- Test Equipment for Use with Electrical and Electronic Equipment, General Specification for.
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-I-46058	- Insulating Compound, Electrical (For Coating Printed Circuit Assemblies).

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-403	- Preparation for and Installation of Rivets and Screws, Rocket and Missile Structure.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461A Notice 4	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 Notice 3	- Electromagnetic Interface Characteristics, Measurement of.
MIL-STD-882	- System Safety Program Requirements
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-45662	- Calibration System Requirements.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ARMY

1231500	- Top Assembly Drawing, SC.
12335633	- Non-Complex Computer Program Configuration Item, Set Com System.
12335819	- Acceptance Test Procedure, SC.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENT

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish Set Coms which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent equipment delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed in and necessary in the construction of this unit, the contractor shall be responsible for using only those products which are listed by part or drawing number as Qualified Products List(s) (QPL) items on the QPL in the referenced specification or which have been approved for inclusion on such lists (see 4.1.4).

3.3 Materials. Materials used by the contractor shall be high quality, suitable for the purpose, and shall conform to applicable Government specifications, referenced drawings, and standards (see 4.9.1 and 6.4).

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3.3.1 Fungistatic materials. All non-metallic materials shall be inherently fungistatic, as defined by requirement 4 of MIL-STD-454, or treated to resist fungus growth, except when used as components of potted or otherwise moisture sealed assemblies (see 4.9.1).

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact unless suitably protected against electrolytic corrosion. The selection and protection of such metal combinations shall be according to requirement 16 of MIL-STD-454 (see 4.9.1).

3.4 Design and construction.

3.4.1 Design, construction, and workmanship. The Set Com shall conform to the applicable requirements of MIL-T-28800 for design, construction, and workmanship, except as otherwise specified herein (see 4.9.2).

3.4.2 Standards of manufacture. Processes and parts used in the construction of the Set Com shall be in accordance with drawings 1231500 and 12335633, parts lists, other documents, and subordinate documentation. All integrated circuits shall be screened in accordance with the requirements specified on the individual envelope drawings for each of the integrated circuits (see 4.9.1).

3.4.3 Item description. The Set Com shall be a display/controller device capable of being the interface between a test set similar to the STE-X set and the operator. A keyboard will provide the means to input test numbers, respond to test set questions and issue commands to control the test flow. Additionally, it shall be capable of performing Digital Multimeter (DMM) functions in a stand-alone mode.

3.4.4 Characteristics.

3.4.4.1 Performance.

3.4.4.1.1 DMM performance. In the DMM mode of operation, the Set Com shall be capable of measuring AC and DC voltage, AC and DC currents, frequency, and resistance. All functions shall be autoranging to minimize external switches and actuators.

3.4.4.1.1.1 Voltage measurements. The Set Com shall be capable of measuring DC and AC peak voltages up to 200 VDC and 140 VAC RMS. Accuracies shall be as follows:

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Function	Measurement Range	Maximum Error Full Scale	Reading
DC Volts	0.000 - 1.999	+/-0.2%	+/-1.0%
	02.00 - 19.99	+/-0.1%	+/-1.2%
	20.00 - 199.9	+/-0.1%	+/-1.2%
AC Volts	0.030 - 1.399	+/-0.3%	+/-1.4%
	1.400 - 13.99	+/-0.3%	+/-1.6%
	14.0 - 140.0	+/-0.3%	+/-1.6%

The frequency range of AC voltage measurements shall be 30 Hz to 500 Hz.

3.4.4.1.1.2 Current measurements. The Set Com shall be capable of measuring DC and AC RMS currents up to 2A for DC and 1.4A for RMS AC currents. Accuracy shall be as follows:

Function	Measurement Range (A)	Maximum Error Full Scale	Reading
DC current	0.000 - 2.000	+/-0.4%	+/-1.5%
AC current	0.000 - 1.400	+/-0.8%	+/-1.8%

The frequency range of AC current measurements shall be 30 Hz to 500 Hz.

3.4.4.1.1.3 Frequency measurements. The Set Com shall be capable of measuring frequencies from 40 Hz minimum to a maximum of 1000 Hz with a voltage level between 1V - 400 Vpp. The accuracy shall be ± 0.1% full scale with a resolution of 0.1 Hz.

3.4.4.1.1.4 Resistance measurements. The Set Com shall be capable of measuring resistances in the range of 0 to 999.9 kohms. The accuracy shall be as follows:

Measurement range	Maximum Error Full Scale (FS)	Reading
0.000 - 0.099 kohms	+/-1.0%	+/-2.0%
0.100 - 1.999 kohms	+/-1.0%	+/-2.0%
2.00 - 19.99 kohms	+/-1.0%	+/-2.0%
20.00 - 999.9 kohms	+/-1.0%	+/-3.0%

3.4.4.1.2 Keyboard description. The keyboard shall consist of twenty keys (see table I). Upon the depression of any key, the ASCII hexadecimal code listed for that key shall be presented to the interface connector. The keyboard shall be expandable to represent alpha/numeric characters and special function keys. The functions listed in table I are representative of Set Com functions when used with a test set.

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TABLE I. Set Com keyboard.

Key	ASCII Hexa-Decimal Codes	Alternate Hexadecimal Codes (see Note 1)	Function
0	30	30	For operator entry of numerical data
1	31	31	For operator entry of numerical data
2	32	32	For operator entry of numerical data
3	33	33	For operator entry of numerical data
4	34	34	For operator entry of numerical data
5	35	35	For operator entry of numerical data
6	36	36	For operator entry of numerical data
7	37	37	For operator entry of numerical data
8	38	38	For operator entry of numerical data
9	39	39	For operator entry of numerical data
10	0A	0D	For test initiation. Also for restart of a test interrupted by a STOP key entry or for operator indication to the test set that a required action has been completed.
STOP	0D	03	For interruption of a test in progress. The Set Com shall display the number code of the test step which is interrupted (see Note 2).
RETEST	2F	52	For rerun of the last test step after a STOP key entry.
CLEAR	20	15	For clearing operator entries prior to GO key entry and for re-establishing a condition for new test entry.

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TABLE I. Set Com keyboard - Continued.

Key	ASCII Hexa-Decimal Codes	Alternate Hexadecimal Codes (see Note 1)	Function
YES	3D	59	For response to test set queries of operator observations.
NO	2E	4E	For response to test set queries of operator observations.
CAL	2A	43	Provides for offset calibration when supported by the test set.
STEP	2D	0A	Set Com Mode: When used with the GO key it permits the operator to manually single step thru a test. DMM Mode: Depressing the STEP key increments the DMM from one measurement function to the next.
↑	---	See Note 3	Provides for back scrolling when supported by the test set. See 3.4.4.1.5 and 3.4.4.1.6.
↓	---	See Note 4	Provides for forward scrolling when supported by the test set. See. 3.4.4.1.5 and 3.4.4.1.6.

Note 1: The alternate character set is invoked by sending the Set Com the following three ASCII hexadecimal messages in the order listed:
1B 23 58

The default character set is invoked by sending the Set Com the following three ASCII hexadecimal messages in the order listed:
1B 23 54

Alternate hexadecimal code display is contingent upon Set Com being interfaced with STE-X or STE-M1/FVS system components and is not a requirement for Set Com Government acceptance testing.

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Note 2: After pushing STOP, the operator has three options:

(1) Hit GO or STEP to continue testing at the beginning of the current test step.

(2) Hit RETEST to continue testing beginning with the last completed test step which invoked the display.

(3) Hit CLEAR to allow a new test number entry.

Note 3: During backward scrolling the Set Com sends the following three ASCII hexadecimal messages in the order listed: 1B 5B 41

and must receive three ASCII hexadecimal messages in the order listed: 1B 5B 53

Note 4: During forward scrolling the Set Com sends the following three ASCII hexadecimal messages in the order listed: 1B 5B 42

and must receive three ASCII hexadecimal messages in the order listed: 1B 5B 52

3.4.4.1.3 Display description. The display shall consist of two rows of thirty-six alpha/numeric characters.

3.4.4.1.4 Interface description. The Set Com shall be capable of supporting a 20 mA serial current interface with selectable baud rates of 9600, 4800, 2400, 1200, 300, 150, and 110.

3.4.4.1.5 Scrolling. In order to accommodate messages of varying lengths (number of lines) the Set Com allows lines to be marked at the beginning of a message. Then, when scrolling is performed, these invisible marks are placed at the upper left corner of the display (message based scrolling). When the Set Com is commanded to scroll down, it will search for the first start of message (SOM) mark before the top line of the display and scroll display so that the line marked is the top line of the display. If no SOMs mark any lines in the Set Com, it will scroll one line. When the Set Com scrolls backward (toward older messages) it stops at the oldest complete message. When the Set Com scrolls forward (toward newer messages) it stops with the display as it was before scrolling commenced.

3.4.4.1.5.1 Scrolling indicator. Whenever the display has been scrolled, all blanks are replaced by a special scroll character indicator to remind the operator that he has scrolled.

3.4.4.1.5.2 Scrolling exit. When the display is scrolled, the keyboard is partially disabled. Only the GO and STOP keys are active, and they are not sent to the test set. They cause the Set Com to immediately leave the scrolling mode.

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3.4.4.1.6 Control characters and sequences. The characters listed in table II and the sequence listed in table III shall be used to control the operation of the Set Com.

3.4.4.1.7 Power requirements. The Set Com shall contain provisions for a power source that shall provide a minimum of six hours of continuous operation in the DMM mode at an ambient temperature of -40°F (-40°C). At a temperature of 70°F (21°C) the Set Com will provide a minimum of thirteen hours of continuous operation in the DMM mode.

3.4.4.2 Physical characteristics.

3.4.4.2.1 Weight. The maximum weight of a single Set Com shall be 2.5 pounds (1.15 kilograms) (see 4.9.3.2.1).

3.4.4.2.2 Dimension. The Set Com shall be a hand held instrument. The maximum dimensions shall be no more than 8.5 inches (216 mm) long by 7.4 inches (188 mm) wide by 2.7 inches (69 mm) (see 4.4.3.2.2).

TABLE II. Control characters.

Name	ASCII Hexadecimal Code	Function
Backspace (BS)	08	Moves the active display position one character to the left.
Line Feed (LF)	0A	Scrolls display one line if the active display position is on the bottom line move the active display position to the left margin of the bottom line.
Carriage Return (CR)	0D	Moves the active display position to the left margin.
Forward Space (FS)	10	Moves the active display position one character to the right.
Cancel (CAN)	18	Cancels any escape sequence in progress.
SUB	1A	Cancels any escape sequence in progress.
Escape (ESC)	1B	Signals that the following characters are special sequences. See table III for a description of these sequences.

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TABLE III. Control sequence.

Name	ASCII Hexadecimal Code	Function
ESC [Pn R	1B, 5B, (Pn), 52	Scroll the display up.
ESC [Pn S	1B, 5B, (Pn), 53	Scroll the display down.
ESC E	1B, 45	Perform a line feed, and mark the current line as the start of a message.
ESC # T	1B, 23, 54	Use default character set (see table I).
ESC # X	1B, 23, 58	Use alternate character set (see table I).
ESC [A	1B, 5B, 41	Move the active display position up one line.
ESC [B	1B, 5B, 42	Move the active display position down one line.
ESC [C	1B, 5B, 43	Move the active display position one character to the right.
ESC [D	1B, 5B, 44	Move the active display position one character to the left.
ESC [3 J	1B, 5B, 33, 4A	Erase the entire scroll memory.
ESC [0 K	1B, 5B, 30, 4B	Erase from the active display position to the end of the line, inclusive.
ESC [1 K	1B, 5B, 31, 4B	Erase from the beginning of the line to the active display position, inclusive.
ESC [2 K	1B, 5B, 32, 4B	Erase the entire line. The active display position is not moved.
ESC Pn {	1B, (Pn), 7B	Change the left margin to Pn. Pn is a decimal number in the range 1 to 36.

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TABLE III. Control sequence - Continued.

Name	ASCII Hexadecimal Code	Function
ESC Pn}	1B, (Pn), 7D	Change the right margin to Pn. Pn is a decimal number in the range 1 to 36.
ESC Pn ~	1B, (Pn), 7E	Change to baud rate according to Pn and the following table: Pn — Baud rate 0 — 110 1 — 150 2 — 300 3 — 1200 4 — 2400 5 — 4800 6 — 9600

3.4.4.3 Reliability. The Set Com shall have a design goal of 4000 hours mean-time-between-failure (MTBF). As a minimum, the Set Com shall have a MTBF of 2000 hours (see 4.9.3.3).

3.4.4.4 Maintainability.

3.4.4.4.1 Maintenance ratio. The maintainability required for the Set Com system, expressed by the maintenance ratio (MR), shall not exceed 0.004. This ratio, which is defined as the ratio of maintenance manhours to operating hours, includes organizational and direct support levels but excludes daily operator checks and services. The Set Com shall require no more than one man hour of unscheduled maintenance per 248 hours of operation (see 4.9.3.4).

3.4.4.4.2 Mean time to repair (MTTR). The MTTR for the Set Com including test, repair and verification times shall be one hour at direct support levels of maintenance (see 4.9.3.4).

3.4.4.4.3 Maximum repair time. The maximum acceptable time for test, repair and verification, for any failure of Set Com at direct support levels of maintenance, is 4 manhours (see 4.9.3.4).

3.4.4.4.4 Preventive maintenance. The Set Com system shall require no preventive maintenance exclusive of operator daily checks and services (see 4.9.3.4).

3.4.4.4.5 Modularity. The Set Com system shall be modular in design with easy access to high failure rate components (see 4.9.3.4).

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3.4.4.5 Environmental.

3.4.4.5.1 High temperature (operating). The Set Com shall be capable of operation without damage or malfunction at a high temperature of 125°F (52°C) (see 4.9.3.5.1).

3.4.4.5.2 Low temperature (operating). The Set Com shall be capable of operation without damage or malfunction at a low temperature of -25°F (-32°C). For ambient temperatures below -25°F (-32°C), the Set Com shall be operated in a shelter (see 4.9.3.5.2).

3.4.4.5.3 High temperature (nonoperating). The Set Com shall be capable of being stored and transported in its case without damage at an ambient temperature of 160°F (71°C) (see 4.9.3.5.1).

3.4.4.5.4 Low temperature (nonoperating). The Set Com shall be capable of being stored and transported in its case without damage in ambient temperatures of -60°F (-51°C) (see 4.9.3.5.2).

3.4.4.5.5 Atmospheric pressure (operating). The Set Com shall be capable of operation without damage or malfunction in an atmospheric pressure of 20.5 to 31.0 inches of mercury (Hg) (620 to 785 mm Hg) (see 4.9.3.5.3).

3.4.4.5.6 Atmospheric pressure (nonoperating). The Set Com shall be capable of being stored and transported in its case without damage in atmospheric pressures of 3.9 to 31.0 inches Hg (110 to 785 mm Hg) (see 4.9.3.5.3).

3.4.4.5.7 Relative humidity. The Set Com shall be capable of operation without damage or malfunction under exposure to relative humidity of as low as 2% or as high as 100% (see 4.9.3.5.4).

3.4.4.5.8 Solar radiation. The Set Com shall be capable of operation without damage or malfunction under exposure to solar radiation from 0 to 360 BTU/hr-ft^2 (0 to 1135 W/m^2) at a maximum ambient temperature of 115°F (46°C) (see 4.9.3.5.5).

3.4.4.5.9 Fungus. The Set Com shall evidence no fungus growth which shall affect its operation on all nonmetallic materials not inherently fungistatic under cyclic temperatures of 77 ± 2°F to 86 ± 2°F (25 ± 1°C to 30 ± 1°C) and humidity conditions of 95 ± 5% to 100% relative humidity (see 4.9.3.5.6).

3.4.4.5.10 Dust (fine sand). The Set Com shall show no evidence of abrasion or malfunction after exposure to dust concentrations of 0.3 ± 0.2 grams per cubic foot (g/ft^3) (10.59 ± 7.06 grams per cubic meter (g/m^3)) at air speeds of 300 ± 200 feet per minute (fpm) to 1750 ± 250 fpm (1.52 ± 1.02 meters per second (m/s) to 8.89 ± 1.27 m/s) relative humidity not to exceed 22% and temperature ranging from 73°F to 145°F (23°C to 63°C) (see 4.9.3.5.7).

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3.4.4.5.11 Salt fog. The Set Com shall show no evidence of corrosion or malfunction after exposure to a 5 ± 1% salt solution atomized at 95°F (35°C) with a pH range of 6.5 to 7.2 (see 4.9.3.5.8).

3.4.4.5.12 Drop shock. The Set Com shall evidence no degradation of function after being dropped from a height of four feet on each face, edge and corner (see 4.9.3.5.9).

3.4.4.5.13 Vibration. The Set Com shall evidence no degradation of function after being subjected to sinusoided cycling along each of three mutually perpendicular axes for 84 minutes per axis at a maximum acceleration of 3.5g (29.42 m/s^2) (see 4.9.3.5.10).

3.4.4.5.14 Rain. The Set Com shall be capable of operation, without damage or malfunctions, when exposed to rain of up to 4 inches (102 mm) per hour (see 4.9.3.5.11).

3.4.4.5.15 Electromagnetic interference susceptibility. The Set Com shall operate as specified by an approved test plan in the electromagnetic environment of an automotive garage in the presence of operating vehicles and arc welders (see 4.9.3.5.12).

3.4.5 Materials and construction.

3.4.5.1 Materials, processes, and parts. The selection of materials, parts, and processes shall ensure that their use is commensurate with the equipment end use (see 4.9.3.6.1).

3.4.5.1.1 Restricted and strategic materials. Restricted and strategic materials as defined in MIL-T-28800 shall not be used (see 4.9.3.6.1.1).

3.4.5.1.2 Selection of materials, processes, and parts. MIL-P-11268 shall be used as a guide in selecting materials, processes and parts for the Set Com. Maximum design freedom shall be allowed in the selection of parts of commercial origin and in the design of new parts when such parts offer marked improvement in design, performance, reliability, reduction in cost, size, and weight. Components already in the Army supply system, which may not be formally designated as standard, shall be used in preference to the creation of a new part.

The following policy shall be employed to the maximum extent practicable in selecting parts for the Set Com (see 4.9.3.6.1.2):

Semi-conductors shall be MIL-Standard (JAN parts)

Resistors and capacitors shall be MIL-Standard establishing reliability parts

Microelectronic devices shall be commercial devices with the following types of 100% screening tests, the minimum extent of which will be specified on the detailed part drawings

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- Stabilization bake
- Temperature cycling
- Burn-in
- Final electrical test after burn-in

3.4.5.1.3 External finish. The external surfaces of the Set Com shall be color impregnated in accordance with color number 24087 of FED-STD-595 unless otherwise specified (see 4.9.3.6.1.3).

3.4.5.1.4 Welding. Welding shall be in accordance with requirement 13 of MIL-STD-454 (see 4.9.3.6.1.4).

3.4.5.1.5 Riveting. Riveting shall be in accordance with MIL-STD-403 (see 4.9.3.6.1.5).

3.4.5.1.6 Soldering. Soldering shall be in accordance with requirement 5 of MIL-STD-454 (see 4.9.3.6.1.6).

3.4.5.1.7 Printed wiring assemblies. The printed wiring assemblies shall conform to requirement 17 of MIL-STD-454, except as modified by the manufacturing drawings (see 4.9.3.6.1.7).

3.4.5.1.7.1 Conformal coating. Printed wiring assemblies not containing cap layers shall be conformally coated with a material that conforms to MIL-I-46058. The board assemblies shall be cleaned of flux and other contaminants before coating. Adjustable components, electrical and mechanical mating surfaces shall not be coated (see 4.9.3.6.1.7.1).

3.4.5.1.8 Wiring and cabling. Insulated wires shall be formed into cables, except where it is physically impractical. Wires and cables shall be properly supported and secured to prevent undue stress on the conductors and terminals. Wiring shall be placed and protected to avoid contact with sharp edges. All wires shall allow sufficient slack for servicing and part replacement. Soldered wire terminations shall be mechanically secured so that the connections are not dependent for strength on solder alone (see 4.9.3.6.1.8).

3.4.5.1.9 Interconnecting cables.

3.4.5.1.9.1 Cables, connectors, and cable assemblies. Cables, connectors, and cable assemblies (interconnecting) shall furnish satisfactory electrical connectors and shall be so arranged and wired that no energized leads are terminated in pins or exposed contacts which might be accidentally shorted The live or hot side of the unmated connectors shall be socket rather than pin type.

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3.4.5.1.9.2 Shielded cables. The shield shall be grounded within the equipment when the cable is an integral part thereof. The shield shall be grounded to the metal shell of the connector when a connector is used on the cable. Except for coaxial cable, a ground lead shall be included in the cable rather than depending on the shield for ground connections. The shield shall be securely bonded to the ground lead at least at one point on the cable in a manner designed to minimize impedance between the shield and the ground lead, except when this requirement conflicts with the electromagnetic interference (EMI) requirements specified in 3.4.5.2.

3.4.5.2 Electromagnetic interference and compatibility (EMI/EMC).

3.4.5.2.1 EMI. The Set Com shall be designed to meet the requirements of MIL-STD-461A Notice 4, part 4, for class A3 equipment (see 4.9.3.6.2.1). All test methods listed below are design goals:

Conducted emissions
CE01 - Power leads, 30 Hz to 15 kHz

Conducted susceptibility
CS01 - Power leads, 30 Hz to 50 kHz
CS02 - Power leads, 50 kHz to 400 MHz

3.4.5.2.2 EMC. The Set Com, when interfaced to the system under test and operating as intended, shall not cause undesirable response or performance degradation of the system under test, and shall not itself exhibit undesirable response or performance degradation when operated through all applicable modes or combined operations (see 4.9.3.6.2.2).

3.4.5.3 Identification and marking. Identification and marking shall be in accordance with the requirements of MIL-STD-130 (see 4.9.3.6.3).

3.4.5.4 Workmanship. Workmanship shall be in accordance with the requirements of MIL-STD-454, requirement 9 (see 4.9.3.6.4).

3.4.5.5 Interchangeability. Set Com interchangeability shall be in accordance with requirement 7 of MIL-STD-454 and as detailed in MIL-T-28800, (see 4.9.3.6.5).

3.4.5.6 MANPRINT

3.4.5.6.1 Safety.

3.4.5.6.1.1 Personnel. The Set Com shall be designed to provide for safety of personnel during normal operation, maintenance, or repair of the Set Com.

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3.4.5.6.1.1.1 Electrical. The Set Com shall be designed to provide for electrical safety of personnel, including fail safe (see 4.9.3.6.6.1.1.1).

3.4.5.6.1.1.2 Mechanical. The Set Com shall be designed to provide for mechanical safety of personnel (see 4.9.3.6.6.1.1.2).

3.4.5.6.2 Human factors, human engineering. The Set Com shall be designed to incorporate human engineering criteria (see 4.9.3.6.7).

3.5 Government-loaned property. The Government-loaned property shall be determined by the contracting officer (see 6.2 and 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment shall meet the requirements of MIL-I-45607 and must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662. The contractor will retain on file calibration records for all test equipment, including the period for which the calibration is valid.

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4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operation to determine that practices, methods, and procedures of the written quality assurance system plan are being properly applied, and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedure, or instances of poor practices which might have an adverse effect upon quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been demonstrated.

4.1.4 Qualified products. When a part or component is specified to conform to a specification having a Qualified Products List (QPL), the contractor shall make available, upon request, to the Government documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity.

4.1.5 Quality assurance provisions. In the conduct of inspection, the contractor shall adhere to Quality Assurance Provisions (QAP) and General Quality Assurance Provision (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.1.6 Certification. Where certification is required to verify material or component conformance to a specification, the contractor shall furnish such certification along with documented test results and performance and analytical data.

4.1.7 Final inspection record (FIR). The contractor shall complete a FIR for each Set Com inspected at the place of manufacture.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).
b. Quality conformance inspections (QCI) (see 4.5).
c. Control test (see 4.6).
d. Comparison tests (see 4.7).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 75 ± 5°F (24 ± 3°C).
b. Barometric pressure: 28.5 (+2.0, -3.0) inches Hg
   (725 (+50, -75) mm Hg).
c. Relative humidity: Less than 60%.

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4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as required in accordance with a Government approved test plan. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply Set Coms that are fully representative of those inspected as a first article sample. Any changes or deiviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Production inspection. When specified (see 6.2), the preproduction sample shall consist of three Set Coms. Preproduction inspection shall consist of inspection as specified in table IV in the order specified in table V.

TABLE IV. Classification of inspection.

Title	Requirement	Inspection	First article sample	QCI	Control	Comparison
Mateials, design, and standards of manufacture	3.3 thru 3.3.2 and 3.4.2	4.9.1	X			
Defects	3.4.1, 3.4.5.3, and 3.5	4.9.2	X	X	X	X
Performance	3.4.4.1 thru 3.4.4.1.7	4.9.3.1	X	X	X	
Physical characteristics	3.4.4.2	4.9.3.2	X	X	X	X
Weight	3.4.4.2.1	4.9.3.2.1	X	X	X	X
Dimensions	3.4.4.2.2	4.9.3.2.2	X	X	X	X
Reliability	3.4.4.3	4.9.3.3	X			
Maintainability	3.4.4.4 thru 3.4.4.4.5	4.9.3.4	X			
High temperature	3.4.4.5.1 3.4.4.5.3	4.9.3.5.1	X		X	
Low temperature	3.4.4.5.2 3.4.4.5.4	4.9.3.5.2	X		X	
Atmospheric pressure	3.4.4.5.5 3.4.4.5.6	4.9.3.5.3	X		X	
Relative humidity	3.4.4.5.7	4.9.3.5.4	X		X	
Solar radiation	3.4.4.5.8	4.9.3.5.5	X			
Fungus	3.4.4.5.9	4.9.3.5.6	X			
Sand and dust	3.4.4.5.10	4.8.3.5.7	X			
Salt atmosphere	3.4.4.5.11	4.9.3.5.8	X			
Drop shock	3.4.4.5.12	4.9.3.5.9	X			
Vibration	3.4.4.5.13	4.9.3.5.10	X			
Rain	3.4.4.5.14	4.9.3.5.11	X			

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TABLE IV. Classification of inspection - Continued.

Title	Requirement	Inspection	First article sample	QCI	Control	Comparison
Electromagnetic interference susceptibility	3.4.4.5.15	4.9.3.5.12	X			
Materials, processes and parts	3.4.5.1	4.9.3.6.1	X		X	
Restricted and strategic materials	3.4.5.1.1	4.9.3.6.1.1	X		X	
Selection of materials, processes and parts	3.4.5.1.2	4.9.3.6.1.2	X		X	
External finish	3.4.5.1.3	4.9.3.6.1.3	X			
Welding	3.4.5.1.4	4.9.3.6.1.4	X			
Riveting	3.4.5.1.5	4.9.3.6.1.5	X			
Soldering	3.4.5.1.6	4.9.3.6.1.6	X			
Printed wiring assemblies	3.4.5.1.7	4.9.3.6.1.7	X			
Conformal coatings	3.4.5.1.7.1	4.9.3.6.1.7.1	X			
Wiring and cabling	3.4.5.1.8	4.9.3.6.1.8	X			
Interconnecting cables	3.4.5.1.9	4.9.3.6.1.9	X			
Electromagnetic interference	3.4.5.2.1	4.9.3.6.2.1	X			
Electromagnetic compatibility	3.4.5.2.2	4.9.3.6.2.2	X			
Identification marking	3.4.5.3	4.9.3.6.3	X			
Workmanship	3.4.5.4	4.9.3.6.4	X			
Interchangeability	3.4.5.5	4.9.3.6.5	X			
Electrical	3.4.5.6.1.1.1	4.9.3.6.6.1.1.1	X			
Mechanical	3.4.5.6.1.1.2	4.9.3.6.6.1.1.2	X			
Human factors/ human engineering	3.4.5.6.2	4.9.3.6.6.2	X			

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TABLE V. Sequence and inspection applied to each sample - Continued.

Test sequence	Paragraph number	Preproduction sample			Initial production sample		
		1	2	3	1	2	3
QCI	4.5	X	X	X	X	X	X
Performance	4.9.3.1	X			X		
Physical characteristics	4.9.3.2 thru 4.9.3.2.2	X	X	X	X	X	X
High temperature	4.9.3.5.1		X	X		X	X
Low temperature	4.9.3.5.2		X	X		X	X
Solar radiation	4.9.3.5.5		X	X		X	X
Atmospheric pressure	4.9.3.5.3		X	X		X	X
Relative humidity	4.9.3.5.4		X	X		X	X
Rain	4.9.3.5.11		X	X		X	X
Sand and dust	4.9.3.5.7		X	X		X	X
Salt atmosphere	4.9.3.5.8		X	X		X	X
Vibration	4.9.3.5.10		X	X		X	X
Electromagnetic interference susceptibility	4.9.3.5.12		X	X		X	X
Drop shock	4.9.3.5.9		X	X		X	X
Fungus	4.9.3.5.6		X	X		X	X
Materials, processes and parts	4.9.3.6.1		X	X		X	X
Restricted and strategic materials	4.9.3.6.1.1		X	X		X	X
Selection of materials, processes and parts	4.9.3.6.1.2		X	X		X	X
External finish	4.9.3.6.1.3		X	X		X	X
Welding	4.9.3.6.1.4		X	X		X	X
Riveting	4.9.3.6.1.5		X	X		X	X
Soldering	4.9.3.6.1.6		X	X		X	X
Printed wiring assemblies	4.9.3.6.1.7		X	X		X	X
Conformal coatings	4.9.3.6.1.7.1		X	X		X	X
Wiring and cabling	4.9.3.6.1.8		X	X		X	X
Interconnecting cables	4.9.3.6.1.9		X	X		X	X
Electromagnetic interference	4.9.3.6.2.1		X	X		X	X
Electromagnetic compatibility	4.9.3.6.2.2		X	X		X	X
Identification and marking	4.9.3.6.3		X	X		X	X
Workmanship	4.9.3.6.4		X	X		X	X
Interchangeability	4.9.3.6.5		X	X		X	X
Electrical	4.9.3.6.6.1.1		X	X		X	X
Mechanical	4.9.3.6.6.1.2		X	X		X	X
Human factors/human engineering	4.9.3.6.7		X	X		X	X

NOTE: Test sequence may be altered to minimize cost or schedule with the exception of vibration and drop shock which are last.

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4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select three Set Coms, from the first 20 Set Coms produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table IV in the order specified in table V.

NOTE: After completion of the first article test, the approved Set Coms shall be retained by the contractor for use as a manufacturing standard until completion of the production run. At that time, they will be refurbished and delivered as specified under the contract.

4.4.3 First article inspection failure. Set Com deficiencies during, or as a result of, the first article test, shall be cause for rejection of the Set Coms until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of the first article test shall be evidence that all Set Coms already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all Set Coms shall be corrected by the contractor. The Government will not accept products until first article testing is completed to the satisfaction of the Government.

4.5 Quality conformance inspections (QCI). QCI shall be performed on Set Coms selected in accordance with 4.5.2.1.2 and subjected to the inspection and test specified in table IV. This shall utilize an approved FIR. The FIR shall be made available to the Government for review upon request.

4.5.1 One hundred percent. Each Set Com shall be subjected to electrical performance testing in accordance with the Government approved acceptance test procedure.

4.5.2 Acceptable quality level. When authorized by the Government contracting authority, one hundred percent QCI shall be waived. Thereafter, sample shall be selected in accordance with 4.5.2.1.2 and shall be inspected to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.1 Sampling.

4.5.2.1.1 Lot formation. An inspection lot shall consist of all the Set Coms produced in one week, submitted at one time for acceptance, or as specified in the contract (see 6.2).

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4.5.2.1.2 Sampling for examination. When sampling is authorized for QCI (see 4.5.2) samples shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall examine at least 20 consecutively produced Set Coms. The examination shall establish that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQL. Examination shall be performed against the classification of defects and AQL specified in table VI.

TABLE VI. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability out of tolerance (see 3.4.4.2.2).	SIE 1/
102	Faulty workmanship affecting function (see 3.4.5.4).	Visual
103	Marking, improper (see 3.4.5.3).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.4.4.2.2).	SIE
202	Faulty workmanship affecting appearance (see 3.4.5.4).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Defects. The following rules shall be applied to the counting of defects.

4.5.3.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period, will be added to the minor defects classification with no increase in AQLs.

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4.5.3.2 Recurring major deficiencies. A major deficiency is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected by the contractor when found.

4.5.3.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies and correction shall be accomplished prior to acceptance.

4.6 Control tests. Control tests shall be conducted on one Set Com, selected by the local Government quality assurance representative, from each lot formation specified in 4.5.2.1.1 and tested to the control test specified in table IV. If the Set Com successfully passes all inspections and tests for that production lot, the control test for the following month's production lot may be waived at the discretion of the Government quality assurance representative, with control tests resuming after that month's production lot.

4.7 Comparison test. The Government may randomly select Set Coms at anytime during the production contract period and subject these Set Coms to the comparison tests specified in table IV. Tests shall be conducted by the Government at a site it shall select.

4.8 Failure. Failure of any Set Com to pass any of the specified inspections or failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government. Such corrective action required to correct defects will be at no charge to the Government.

4.9 Methods of inspection.

4.9.1 Materials, design, and standards of manufacture. Conformance to 3.3 through 3.3.2 and 3.4.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, applicable QARs, QAPs, and SQAPs, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.9.2 Defects. Conformance to 3.4.1, 3.4.5.3, and 3.5 shall be determined by examination for the defects listed in table VI. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.9.3 Characteristics.

4.9.3.1 Performance. A quality conformance inspection shall be performed in accordance with table IV. Unless otherwise specified, Set Com performance testing will be conducted at 75 ± 5°F (23 ± 3°C) ambient air temperature and less than 60% relative humidity; nonoperating environmental tests shall be performed with the Set Com stored in the appropriate transit case with the covers installed and latched. The Set Com shall be tested in accordance with the Government approved Acceptance Test Procedures (Army Drawing No. 12335819) to determine conformance to 3.4.4.1.1 through 3.4.4.1.7. The Set Com performance may be validated with the Set Com in either the DMM (stand alone) mode or hooked up to a STE (STE-X, STE-M1/FVS) system via the Set Com cable. If required, an external power source of 24 ± 6 Vdc, at 15 amperes minimum, shall be used. Connections to external power supply, or any test fixtures or equipment shall be made with the available test set cables, as specified by the applicable test procedure.

4.9.3.2 Physical characteristics. The Set Com shall be inspected for conformance to 3.4.4.2.

4.9.3.2.1 Weight. The Set Com shall be weighed to determine conformance to 3.4.4.2.1.

4.9.3.2.2 Dimensions. The Set Com shall be subjected to a visual and mechanical inspection to determine conformance to 3.4.4.2.2.

4.9.3.3 Reliability. The MTBF requirements specified in 3.4.4.3 for the assembly shall be verified by analysis of test and failure data including consideration of functional and environmental stress levels which affect the failure rate. Use of test and failure data on similar components is permitted providing allowances are made for differences in design, usage, and environments. In the event that test history is not sufficient to provide satisfactory evidence of the assemblies MTBF compliance with the reliability requirement, compliance may be determined on the basis of formal reliability predictions employing recognized and approved failure rate data sources and prediction procedures.

4.9.3.4 Maintainability. To determine conformance to 3.4.4.4, the degree of maintainability achieved shall be determined by design review, review of LSA data, and by review and assessment of test data.

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4.9.3.5 Environmental conditions. Prior to environmental testing, the Set Com shall be inspected and operated under standard ambient conditions to obtain data for determining satisfactory operation during and after environmental testing. Unless otherwise specified, the Set Com shall be installed in the test facility in a manner that will simulate service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. Mechanical or electrical connections not used, but normally protected in service, shall be adequately covered. When installing the Set Com in the test chamber, the chamber shall be at ambient conditions. After installation, the Set Com shall be operated to determine proper installation. After each environmental test, the Set Com shall be inspected and operated and the data compared to that obtained before environmental testing began.

4.9.3.5.1 High temperature. To determine conformance to 3.4.4.5.1 and 3.4.4.5.3, the Set Com shall be subjected to the following test. With relative humidity not in excess of 15%, the Set Com, in its case, shall be maintained at 160°F (71°C) for a period of 48 hours. At the end of 48 hours, the temperature shall be lowered to 125°F (52°C) and the Set Com removed from its case and all hookups made. The Set Com shall be operated and results obtained. With the Set Com nonoperating, the Set Com shall be returned to ambient conditions.

4.9.3.5.2 Low temperature. To determine conformance to 3.4.4.5.2 and 3.4.4.5.4, the Set Com shall be subjected to the following test. With the Set Com in its case, the temperature shall be lowered to -60°F (-51°C) and maintained for 24 hours. At the end of 24 hours, the temperature shall be raised to -25°F (-32°C) and the Set Com removed from its case and all hookups made. The Set Com shall be operated until stabilized and results obtained. With the Set Com nonoperating, the Set Com shall be returned to ambient conditions.

4.9.3.5.3 Atmospheric pressure. To determine conformance to 3.4.4.5.5 and 3.4.4.5.6, the Set Com shall be subjected to the following test. The test chamber pressure shall be reduced at a rate not to exceed 2000 feet per minute (ft/min) (10.2 meters per second (m/s)) to 3.9 inches of Hg (110 mm Hg) and maintained for not less than 1 hour. After 1 hour, the chamber pressure shall be raised to 20.5 inches of Hg (620 mm Hg) at a rate not to exceed 2000 ft/min (10.2 m/s) and maintained not less than 1 hour. The Set Com shall be operated and results obtained. With the Set Com nonoperating, the chamber pressure shall be returned to ambient at a rate not to exceed 2000 ft/min (10.2 m/s).

4.9.3.5.4 Relative humidity. To determine conformance to 3.4.4.5.7, the Set Com shall be subjected to the following test:

a. The test chamber shall be at ambient temperature and humidity.
b. Over a period of 2 hours, the temperature shall be raised to 149°F and the relative humidity raised to 95 +5, -3%. Maintain these conditions for not less than 6 hours.

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c. Reduce the temperature to 86°F in 16 hours while maintaining 85% or greater relative humidity.
d. Repeat b and c for a total of 10 cycles (not less than 240 hours).
e. At the end of the tenth cycle, maintain 86°F and 85% relative humidity. The Set Com shall be operated and results obtained.
f. Return the chamber to standard ambient conditions.

4.9.3.5.5 Solar radiation. To determine conformance to 3.4.4.5.8, the Set Com shall be subjected to the following test. The Set Com shall be exposed to radiant energy at the rate of 104 ± 4 watts per square foot for 48 hours, during which time the chamber temperature shall be maintained at 115°F. At the conclusion of the 48 hour period, with the chamber temperature maintained, the Set Com shall be operated and results obtained. The Set Com shall then be returned to ambient conditions.

4.9.3.5.6 Fungus. To determine conformance to 3.4.4.5.9, the Set Com shall be tested using the apparatus of 4.9.3.5.6.1 and the procedure of 4.9.3.5.6.2.

4.9.3.5.6.1 Apparatus. The apparatus required to conduct this test consists of chambers or cabinets together with auxiliary instrumentation capable of maintaining the specified condition of temperature and humidity. Provisions shall be made to prevent condensation from dripping on the Set Coms. There shall be free circulation of air around the Set Coms and the contact area of fixtures supporting the Set Coms shall be kept to a minimum.

When forced air is employed, the flow should not exceed 1 meter per second over the surface of the test specimen.

4.9.3.5.6.2 Procedure.

4.9.3.5.6.2.1 Preparation of mineral-salts solution. The solution shall contain the following:

Potassium dihydrogen orthophosphate (KH_2PO_4)	0.7 g
Potassium monohydrogen orthophosphate (K_2HPO_4)	0.7 g
Magnesium sulfate heptahydrate ($MgSO_4$ $7H_2O$)	0.7 g
Ammonium nitrate (NH_4NO_3)	1.0 g
Sodium chloride (NaCl)	0.005 g

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Ferrous sulfate heptahydrate ($FeSO_4$ $7H_2O$)	0.002 g
Zinc sulfate heptahydrate ($ZnSO_4$ $7H_2O$)	0.002 g
Manganous sulfate monohydrate ($MnSO_4$ H_2O)	0.001 g
Distilled water	1000 ml

Sterilize the mineral salts solution by autoclaving at 121°C (250°F) for 20 minutes. Adjust the pH of the solution by the addition of 0.01 normal soultion of NaOH so that after sterilization the pH is between 6.0 and 6.5. Prepare sufficient salts solution for the required tests.

a. Reagent grade chemicals shall be used in all tests.

b. Unless otherwise specified, references to water shall be understood to mean distilled water or water of equal purity.

4.9.3.5.6.2.2 Preparation of mixed spore suspension. The following test fungi shall be used:

Fungi	ATCC No. 1/	QM(NLABS) No. 2/
Aspergillus niger	9642	386
Aspergillus flavus	9643	380
Aspergillus versicolor	11730	432
Penicillium funiculosum	11797	474
Chaetomium globosum	6205	459

1/ American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852

2/ Pioneering Research Division, U.S. Army Natick Laboratories, Natick, Massachusetts 01760

Maintain cultures of these fungi separately on an appropriate medium such as potato dextrose agar. However, the culture of chaetomium globosum shall be cultured on strips of filter paper on the surface of mineral salts agar. (Mineral salts agar is identical to mineral salts solution described in 4.9.3.5.6.2.1, but contains in addition 15.0 g of agar per liter). The stock cultures may be kept for not more than 4 months at 6° ± 4°C (43° ± 7°F) at which time subcultures shall be made and new stocks shall be

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selected from the subcultures. If genetic or physiological changes occur, obtain new cultures as specified above. Subcultures used for preparing new stock cultures or the spore suspension shall be incubated at 30°C (86°F) for 9 to 12 days or longer. Prepare a spore suspension of each of the five fungi by pouring into one subculture of each fungus a 10-ml portion of a sterile solution containing 0.05 g per liter of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate or sodium lauryl sulfate. Use a sterile platinum or nichrome inoculating wire to scrape gently the surface growth from the culture of the test organism. Pour the spore charge into a sterile 125-ml glass-stoppered Erlenmeyer flask containing 45 ml of sterile water and 50 to 75 solid glass beads, 5 mm in diameter. Shake the flask vigorously to liberate the spores from the fruiting bodies and to break the spore clumps. Filter the dispersed fungal spore suspension, through a 6 mm layer of glass wool contained in a glass funnel, into a sterile flask. This process should remove large mycelial fragments and clumps of agar which could interfere with the spraying process. Centrifuge the filtered spore suspension aseptically, and discard the supernatant liquid. Resuspend the residue in 50 ml of sterile water and centrifuge. Wash the spores obtained from each of the fungi in this manner three times. Dilute the final washed residue with sterile mineral-salts solution in such a manner that the resultant spore suspension shall contain 1,000,000 ± 200,000 spores per ml as determined with a counting chamber. Repeat this operation for each organism used in the test and blend equal volumes of the resultant spore suspension to obtain the final mixed spore suspension. The spore suspension may be prepared fresh each day or may be held at 6° ± 4°C (43° ± 7°F) for not more than 7 days.

4.9.3.5.6.2.3 Viability of inoculum control. With each daily group of tests, place each of three pieces of sterilized filter paper, 1-inch square, on hardened mineral-salts agar in separate Petri dishes. Inoculate these with the spore suspension by spraying the suspension from a sterilized atomizer 1/ until initiation of droplet coalesence. Incubate these at 30°C (86°F) at a relative humidity not less than 85 percent and examine them after 7 days of incubation. There shall be copious growth on all three of the filter paper control specimens. Absence of such growth requires repetition of the test.

4.9.3.5.6.2.4 Control items. In addition to the viability of inoculum control, known susceptible substrates shall be inoculated along with the test item to insure that proper conditions are present in the incubation chamber to promote fungus growth. The control items shall consist of cotton duct 8.25-ounce strips that are 1.25 inches (3.2 cm) wide, that have been dipped into a solution containing 10 percent glycerol, 0.1 percent potassium dihydrogen orthophosphate (KH_2PO_4), 0.1 percent ammonium nitrate (NH_4NO_3), 0.025 percent magnesium sulfate ($MgSO_4$ $7H_2O$), and 0.05 percent yeast extract (pH 5.3), and from which the excess liquid has been removed. The strips should be hung to air dry before being inoculated and placed into the chamber.

1/ An atomizer capable of providing 15,000 ± 3,000 spores per square centimeter.

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4.9.3.5.6.2.5 Inoculation of test and control item.

a. Mount the test control items on suitable fixtures or suspend from hangers. No cleaning of the test item shall be permitted for 72 hours prior to the beginning of the fungus test. Equipment handling prior to and during the fungus test shall be accomplished without contamination of the equipment.

b. Precondition the chamber and its contents at 30°C (86°F) and 97 ± 2 percent relative humidity for at least 4 hours.

c. Inoculate the test and control items with the mixed fungus spore suspension of 4.9.3.5.6.2.2 by spraying it on and into the test and control items (if not hermetically sealed) in the form of a fine mist from a previously sterilized atomizer or nebulizer. In spraying the test and control items, care should be taken to spray all surfaces which are exposed during use or maintenance. If the surfaces are nonwetting, spray until initiation of droplet coalescence. Incubation is to be started immediately following the inoculation.

4.9.3.5.6.2.6 Incubation of test items.

a. Incubate test items under cyclic temperature and humidity conditions to include 20 hours of relative humidity at 95 ± 5 percent at an air temperature of 30° ± 1°C (86° ± 2°F) followed by 4 hours of 100 percent relative humidity at 25° ± 1°C (77° ± 2°F).

b. After 7 days, inspect the growth on the control items to be assured that the environmental conditions are suitable for growth. If inspection reveals that the environmental conditions are unsuitable for growth, the entire test shall be repeated.

c. If the control items show satisfactory fungus growth, continue the test for a period of 28 days from the time of inoculation.

4.9.3.5.6.2.7 Inspection. At the end of the incubation period, the Set Coms shall be removed from the test chamber and inspected in accordance with 4.9.3.5, except for hermetically sealed equipment, the equipment enclosure shall be opened and the interior examined for evidence of fungus growth or damage. The Set Coms shall be operated and the results obtained in accordance with 4.9.3.5.

NOTE: Conductive solutions used as a spore-media and growth accelerator may affect operational tests.

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4.9.3.5.7 Sand and dust. To determine conformance to 3.4.4.5.10, the Set Com shall be tested using the apparatus of 4.9.3.5.7.1 and the procedure of 4.9.3.5.7.2.

4.9.3.5.7.1 Apparatus. The test facility shall consist of a chamber and accessories to control dust concentration, velocity, temperature, and humidity of dust laden air. In order to provide adequate circulation of the dust laden air, no more than 50 percent of the cross-sectional area (normal to air flow) and 30 percent of the volume of the chamber shall be occupied by the Set Coms. The chamber shall be provided with a suitable means of maintaining and verifying the dust concentration in circulation. A minimum acceptable means for doing this is by use of a properly calibrated smoke meter and standard light source. The dust laden air shall be introduced into the test space in such a manner as to allow it to become approximately laminar in flow before it strikes the Set Com.

4.9.3.5.7.1.1 Dust requirements. The dust used in this test shall be a fine sand (97-99 percent by weight SiO_2) of angular structure, and shall have the following size distribution as determined by weight, using the U.S. Standard Sieve Series:

a. 100 percent of this dust shall pass through a 100-mesh screen.

b. 98 ± 2 percent of the dust shall pass through a 140-mesh screen.

c. 90 ± 2 percent of the dust shall pass through a 200-mesh screen.

d. 75 ± 2 percent of the dust shall pass through a 325-mesh screen.

NOTE: 140-mesh silica flour as produced by the Ottawa Silica Company, Ottawa, Illinois, or equal, is satisfactory for use in the performance of these tests.

4.9.3.5.7.2 Procedure. Prepare the Set Com in accordance with 4.9.3.5, positioning the Set Com as near the center of the chamber as practicable. If more than one Set Com is being tested, there shall be a minimum clearance of 4 inches between surfaces of Set Coms or any other material or object capable of furnishing protection. Also, no surface of the Set Coms shall be closer than 4 inches from any wall of the test chamber. Orient the Set Coms so as to expose the most critical or vulnerable parts to the dust stream.

Step 1 - Set the chamber controls to maintain an internal chamber temperature of 23°C (73°F) and a relative humidity of less than 22 percent. Adjust the air velocity to 1,750 ± 250 ft/min (8.8 ± 1.2 m/s) per minute. Adjust the dust feeder to control the dust concentration at 0.3 ± 0.2 grams per cubic foot. With the Set Com nonoperating, maintain these conditions for 6 hours.

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Step 2 - Stop the dust feed and reduce the air velocity to 300 ± 200 ft/min (1.5 ± 1 m/s). Raise the internal chamber air temperature to 63°C (145°F). Hold these conditions 16 hours.

Step 3 - While holding chamber temperature at 63°C (145°F) adjust the air velocity to 1,750 ± 250 ft/min (8.8 ± 1.2 m/s). Adjust the dust feeder to control the dust concentration at 0.3 ± 0.2 gms per cubic foot. Unless otherwise specified, with the Set Com non-operating, maintain these conditions for 6 hours.

Step 4 - Turn off all chamber controls and allow the Set Com to return to standard ambient conditions Remove accumulated dust from the Set Com by brushing, wiping, or shaking, care being taken to avoid introduction of additional dust into the Set Com. Dust shall not be removed by either air blast or vacuum cleaning.

Step 5 - Operate and inspect the Set Com in accordance with 4.9.3.5.

Step 6 - Inspect the Set Com and obtain results as specified in 4.9.3.5.

4.9.3.5.8 Salt atmosphere. To determine conformance to 3.4.4.5.11, the Set Com shall be tested using the apparatus of 4.9.3.5.8.1 and the procedure of 4.9.3.5.8.2.

4.9.3.5.8.1 Apparatus. The apparatus used in the salt fog test shall include the following:

a. Exposure chamber with racks for supporting Set Coms.

b. Salt solution reservoir with means for maintaining an adequate level of solution.

c. Means for atomizing salt solution, including suitable nozzles and compressed air supply.

d. Chamber heating means and control.

e. Means for humidifying the air at a temperature above the chamber temperature.

4.9.3.5.8.1.1 Chamber. The chamber and all accessories shall be made of material that will not affect the corrosiveness of the fog such as glass, hard rubber, plastic, or kiln dried wood other than plywood. In addition, all parts which come in contact with the Set Com shall be of materials that will not cause electrolytic corrosion. The chamber and accessories shall be constructed and arranged so that there is no direct impingement of the fog or dripping of the condensate on the Set Coms that the fog circulates freely about all Set Coms to the same degree, and that no liquid which has come in

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contact with the Set Coms returns to the salt-solution reservoir. The chamber shall be properly vented to prevent pressure buildup and allow uniform distribution of salt fog. The discharge end of the vent shall be protected from strong drafts which can create strong air currents in the test chamber.

4.9.3.5.8.1.2 Atomizers. The atomizers used shall be of such design and construction as to produce a finely divided, wet, dense fog. Atomizing nozzles shall be made of material that is nonreactive to the salt solution.

4.9.3.5.8.1.3 Air supply. The compressed air entering the atomizer shall be essentially free from all impurities, such as oil and dirt. Means shall be provided to humidify and warm the compressed air as required to meet the operating conditions. The air pressure shall be suitable to produce a finely divided dense fog with the atomizer or atomizers used. To insure against clogging the atomizers by salt deposition, the air should have a relative humidity of at least 85 percent at the point of release from the nozzle. A satisfactory method is to pass the air in very fine bubbles through a tower containing heated water which should be automatically maintained at a constant level. The temperature of the water should be at least 35°C (95°F). The permissible water temperature increases with increasing volume of air and with decreasing insulation of the chamber and the chamber's surroundings. However, the temperature should not exceed a value above which an excess of moisture is introduced into the chamber (for example 43°C (109°F) at an air pressure of 12 psi) or a value which makes it impossible to meet the requirements for operating temperature.

4.9.3.5.8.1.4 Preparation of salt solution. The salt used shall be sodium chloride containing on the dry basis not more than 0.1 percent sodium iodide and not more than 0.5 percent of total impurities. Unless otherwise specified, a 5 ± 1 percent solution shall be prepared by dissolving five parts by weight of salt in 95 parts by weight of distilled or demineralized water. The solution shall be adjusted to and maintained at a specific gravity between the limits shown on figure 1 by utilizing the measured temperature and density of the salt solution. Sodium tetraborate (common borax) may be added to the salt solution in a ratio not to exceed 0.7 gms (1/4 level teaspoon) sodium tetraborate to 20 gallons of salt solution as a pH stabilization agent. Adjustment of pH of the salt solution shall be so maintained that the solution atomized at 35°C (95°F) and collected by the method specified in 4.9.3.5.8.2.3 will be in the pH range of 6.5 to 7.2. Only diluted C.P. hydrochloric acid or C. P. sodium hydroxide shall be used to adjust the pH. The pH measurement shall be made electrometrically, using a glass electrode with a saturated potassium chloride bridge, or by a colorimetric method, such as bromothymol blue, provided the results are equivalent to those obtained with the electrometric method. The pH shall be measured when proparing each new batch of solution and as specified in 4.9.3.5.8.2.4.

4.9.3.5.8.1.5 Filter. A filter fabricated of noncorrosive materials similar to that shown in figure 2 shall be provided in the supply line and immersed in the salt solution reservoir in a horizontal manner.

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4.9.3.5.8.2 Procedure.

4.9.3.5.8.2.1 Temperature. The test shall be conducted with a temperature in the exposure zone maintained at 35°C (95°F). Satisfactory methods for controlling the temperature accurately are by housing the apparatus in a properly controlled constant temperature room, by thoroughly insulating the apparatus and preheating the air to the proper temperature prior to atomization, or by jacketing the apparatus and controlling the temperature of the water or of the air used in the jacket. The use of immersion heaters within the chamber for the purpose of maintaining the temperature within the exposure zone is prohibited.

4.9.3.5.8.2.2 Atomization. Suitable atomization has been obtained in chambers having a volume of less than 12 cubic feet under the following conditions:

a. Nozzle pressure shall be as low as practicable to produce fog at the required rate.

b. Orifices between 0.02 and 0.03 inch in diameter.

c. Atomization of approximately 3 quarts of salt solution per 10 cubic feet of chamber volume per 24 hours.

When using large size chambers having a volume considerably in excess of 12 cubic feet, the conditions specified may require modification to meet the requirements for operating conditions.

4.9.3.5.8.2.3 Placement of salt fog collection receptacles. The salt fog conditions maintained in all parts of the exposure zone shall be such that a clean fog collecting receptacle placed at any point in the exposure zone will collect from 0.5 to 3 milliliters of solution per hour for each 80 square centimeters of horizontal collecting area (10 centimeters diameter) based on an average test of at least 16 hours. A minimum of two receptacles shall be used, one placed nearest to any nozzle and one farthest from all nozzles. Receptacles shall be placed so that they are not shielded by the Set Coms and so no drops of solution from the Set Coms or other sources will be collected.

4.9.3.5.8.2.4 Measurement of salt solution. The solution, collected in a manner specified in 4.9.3.5.8.2.3 shall have the sodium chloride content and pH specified in 4.9.3.5.8.1.4 when measured at a temperature of 35°C (95°F). The salt solution from all collection receptacles used can be combined to provide that quantity required for the measurements specified. Measurements shall be made after each test using the following procedures:

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a. The solution, maintained at the specified temperature, can be measured in a graduate of approximately 2.5 centimeters inside diameter. A small laboratory type hydrometer will be required for measurement within this volume.

b. The pH shall be measured as specified in 4.9.3.5.8.1.4.

NOTE: For salt fog chambers that are used infrequently, a 24-hour test run shall be accomplished followed by the measurements. The Set Com shall not be exposed to this test run.

4.9.3.5.8.2.5 Preparation of Set Com. The Set Com shall be given a minimum of handling, particularly on the significant surfaces, and shall be prepared for test immediately before exposure. Unless otherwise specified, uncoated metallic or metallic coated devices shall be thoroughly cleaned of oil, dirt, and grease as necessary until the surface is free from water break. The cleaning methods shall not include the use of corrosive solvents nor solvents which deposit either corrosive or protective films, nor the use of abrasives other than a paste of pure magnesium oxide. Those portions of the Set Com which come in contact with the support shall be protected with a suitable coating of wax or similar substance impervious to moisture.

4.9.3.5.8.2.6 Performance of test. The Set Com shall be placed in the test chamber in accordance with 4.9.3.5, and exposed to the salt fog for a period of 48 hours. At the end of the exposure period, the Set Com shall be operated and the results compared with the data obtained in accordance with 4.9.3.5. The Set Com shall be inspected for corrosion in accordance with 4.9.3.5. If necessary to aid in examination, a gentle wash in running water no warmer than 38°C (100°F) may be used. The Set Com shall then be stored in an ambient atmosphere for 48 hours. At the end of the drying period, when specified, the Set Com shall be again operated and the results compared with the data obtained in accordance with 4.9.3.5. The Set Com shall then be inspected in accordance with 4.9.3.5.

4.9.3.5.9 Drop shock. To determine conformance to 3.4.4.5.12, the Set Com shall be dropped eight times, once on each face, and once in each of two diagonally opposite corners. The Set Com shall be dropped without its case. Drops shall be made from a quick release hook or drop tester onto a barrier constructed of 2-inch plywood backed by concrete. The Set Com shall be so oriented that upon impact a line from the struck surface or corner to the center of gravity of the Set Com is perpendicular to the impact surface. The Set Com shall be operating during one of the eight drops and the Set Com self test shall be performed after each drop.

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4.9.3.5.10 Vibration. To determine conformance to 3.4.4.5.13, the Set Com shall be rigidly mounted and vibrated along each axis in accordance with the time, frequency, and level as specified in table VII, figure 3, and figure 4.
The Set Com self test shall be made after vibration along each axis.

TABLE VII. Time schedule for vibration test.

Test	Sweep Time
Sinusoidal, cycling for 84 minutes per axis	12 minutes 5-200-5 Hz

4.9.3.5.11 Rain. To determine conformance to 3.4.4.5.14, the Set Com shall be subjected to the following test. A chamber having the capability of producing falling rain shall be used. The chamber temperature shall be uncontrolled throughout the test period. The rain shall be produced by a water distribution device of such design that the water is emitted in the form of droplets having a diameter range between 0.5 and 4.5 millimeters 1/. The temperature of the water shall be between 52 and 95°F. The water distribution shall be such that the rain is dispersed completely over the Set Com. No rust or corrosive contaminants shall be imposed on the Set Com by the test facility.

a. Place the Set Com in the chamber in accordance with 4.9.3.5 and orient in such a manner as to allow impingement of the rain on the Set Com normal to its horizontal axis.

b. For not less than one hour, produce a simulated falling rain in the test chamber stabilized at a constant rate of 4.0 ± 0.5 inches (100 ± 12 mm) per hour.

c. Reorient the Set Com to an angle 45° from the horizontal plane of step a and the index the item around its center axis periodically to allow exposure to rainfall of all the significant surfaces of the test item.

d. Expose the Set Com to the simulated falling rain for a total time of not less than 2 hours, which shall represent the sum of equal exposure periods in the orientation modes of steps a and c.

e. At the conclusion of the rain exposure, remove the Set Com from the chamber. Sponge or lightly wipe accumulated moisture from exterior surfaces and visually inspect the Set Com for compliance. Operate the Set Com in accordance with 4.9.3.5

1/ A recommended method of measuring raindrop size is the flour pellet method as referenced in "The Relation of Raindrop Size to Intensity" by Laws and Parsons, Transactions of the American Geophysical Union, Part II, pps 452 to 459, -1943.

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4.9.3.5.12 Electromagnetic interference susceptibility. To determine conformance to 3.4.4.5.15, the Set Com shall be tested in accordance with the applicable provisions of MIL-E-6051.

4.9.3.6 Materials and conatruction.

4.9.3.6.1 Materials, processes, and parts.

4.9.3.6.1.1 Restricted and strategic materials. To determine conformance to 3.4.5.1, the materials used in the Set Com shall be screened.

4.9.3.6.1.2 Selection of materials, processes, and parts. To determine conformance to 3.4.5.1.2, all materials, processes, and parts shall be inspected.

4.9.3.6.1.3 External finish. To determine conformance to 3.4.5.1.3, a visual inspection shall be made.

4.9.3.6.1.4 Welding. To determine conformance to 3.4.5.1.4, the welds shall be inspected.

4.9.3.6.1.5 Riveting. To determine conformance to 3.4.5.1.5, riveting shall be inspected.

4.9.3.6.1.6 Soldering. To determine conformance to 3.4.5.1.6, soldering shall be inspected.

4.9.3.6.1.7 Printed wiring assemblies. To determine conformance to 3.4.5.1.7, printed wiring assemblies shall conform by design.

4.9.3.6.1.7.1 Conformal coating. To determine conformance to 3.4.5.1.7.1, conformal coating shall be inspected.

4.9.3.6.1.8 Wiring and cabling. To determine conformance to 3.4.5.1.8, wiring and cabling shall be inspected.

4.9.3.6.1.9 Interconnecting cables. To determine conformance to 3.4.5.1.9, interconnecting cables and wire shall be inspected.

4.9.3.6.2 Electromagnetic interference and compatibility (EMI/EMC).

4.9.3.6.2.1 EMI. To determine conformance to 3.4.5.2.1, the measurement equipment setup, procedures, and operations of the test sample shall be in accordance with MIL-STD-462, Notice 3.

4.9.3.6.2.2 EMC. To determine conformance to 3.4.5.2.2, the Set Com and military vehicle compatibility test shall be performed in accordance with MIL-STD-462, Notice 3, system/subsystem compatibility tests.

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4.9.3.6.3 Identification and marking. To determine conformance to 3.4.5.3, the Set Com shall be visually examined to verify product marking.

4.9.3.6.4 Workmanship. To determine conformance to 3.4.5.4, examination of workmanship shall be performed on the assembly and at all phases of assembly and test.

4.9.3.6.5 Interchangeability. To determine conformance to 3.4.5.5, the Set Com shall be inspected.

4.9.3.6.6 MANPRINT.

4.9.3.6.6.1 Safety.

4.9.3.6.6.1.1 Personnel.

4.9.3.6.6.1.1.1 Electrical. To determine conformance to 3.4.5.6.1.1.1, the Set Com design shall be reviewed and inspected to determine compliance with the applicable requirements of MIL-STD-454, requirement 1, and MIL-STD-882.

4.9.3.6.6.1.1.2 Mechanical. To determine conformance to 3.4.5.6.1.1.2, the Set Com design shall be reviewed and inspected to determine compliance with the applicable requirements of MIL-STD-454, section 8, requirement 1.

4.9.3.6.6.2 Human factors/human engineering. To determine conformance to 3.4.5.6.2, operational and maintenance tasks shall be inspected to insure that a satisfactory soldier machine interface has been achieved per MIL-STD-1472.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The test equipment covered by this specification is intended to be used in conjunction with STE-X or STE-M1/FVS (or any future STE system) to diagnose state of readiness or faulty conditions in complex military systems, such as combat and tactical vehicles, howitzers, helicopters, and missile systems, in their "as is condition". The Set Com may also be used as a DMM to make simple electrical measurements.

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6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Military part number.
c. If first article samples are not required (see 3.1).
d. Specify the Government-loaned property (see 3.5).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
g. If inspection conditions shall be other than as specified (see 4.3).
h. If preproduction inspection is required (see 4.4.1).
i. If initial production inspection is not required (see 4.4.2).
j. If inspection lot formation is other than as specified (see 4.5.2.1.1).
k. Selection of applicable level of packaging requirements (see 5.1).
l. If color impregnation is to be other than as specified (see 3.4.5.1.3).

6.3 Inspection for first article. When a first article inspection is required, the first article should consist of three units. One unit should be used for performance verification, and two units should be used for environmental verification.

6.4 Recycled materials. The use of recycled materials that meet the requirements of the applicable material specifications without jeopardizing the intended use of the item should be encouraged (see 3.3).

6.5 Supersession data. This specification supersedes the critical item developement specification for set communicator, specification number 12335887, dated 30 July 1986.

6.6 Government-loaned property. the contracting officer should arrange to loan the property listed in 3.5.

6.7 Subject term (key word) listing.

Display controller
Keyboard
Simplified Test Equipment (STE)
STE-Expandable (STE-X)
STE-Internal Combustion Engine (STE-ICE)
STE-ICE/Reprogrammable (STE-ICE/R)
STE-M1/FVS

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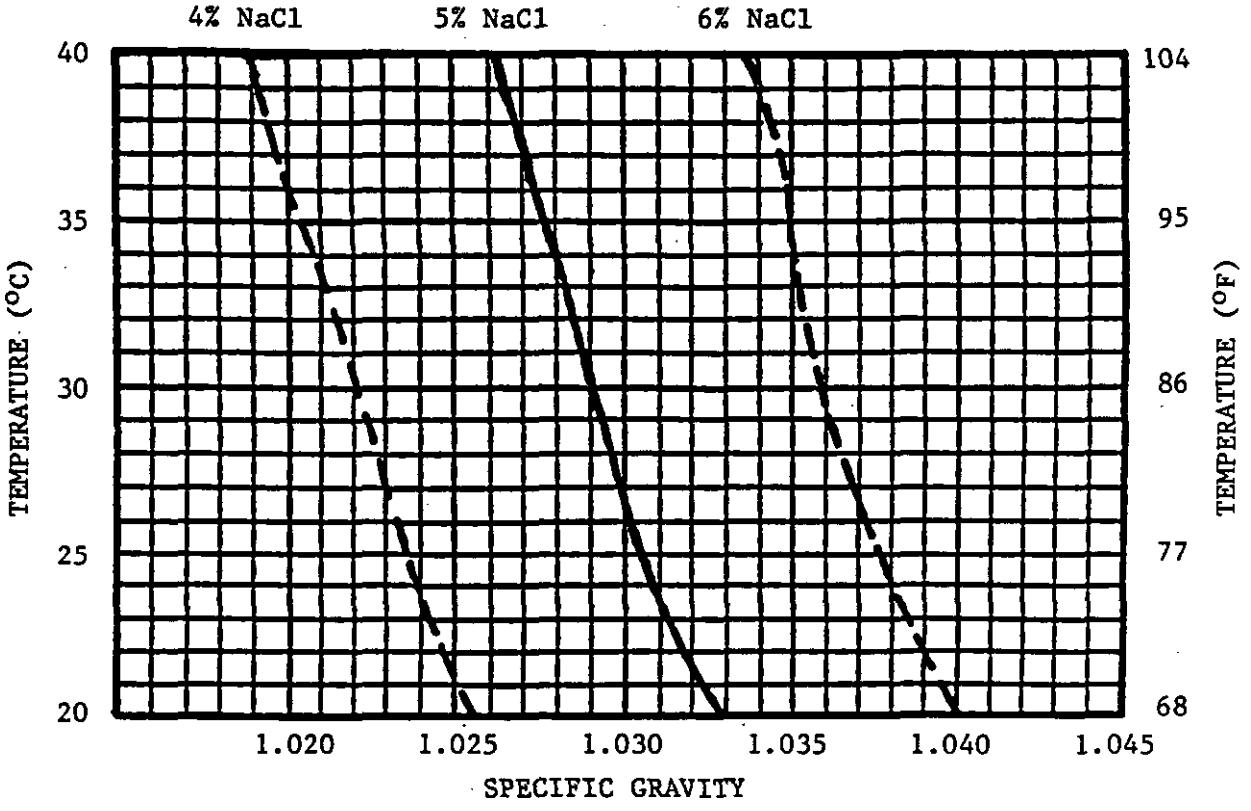


FIGURE 1. Variations of specific gravity of salt (NaCl) solution with temperature.

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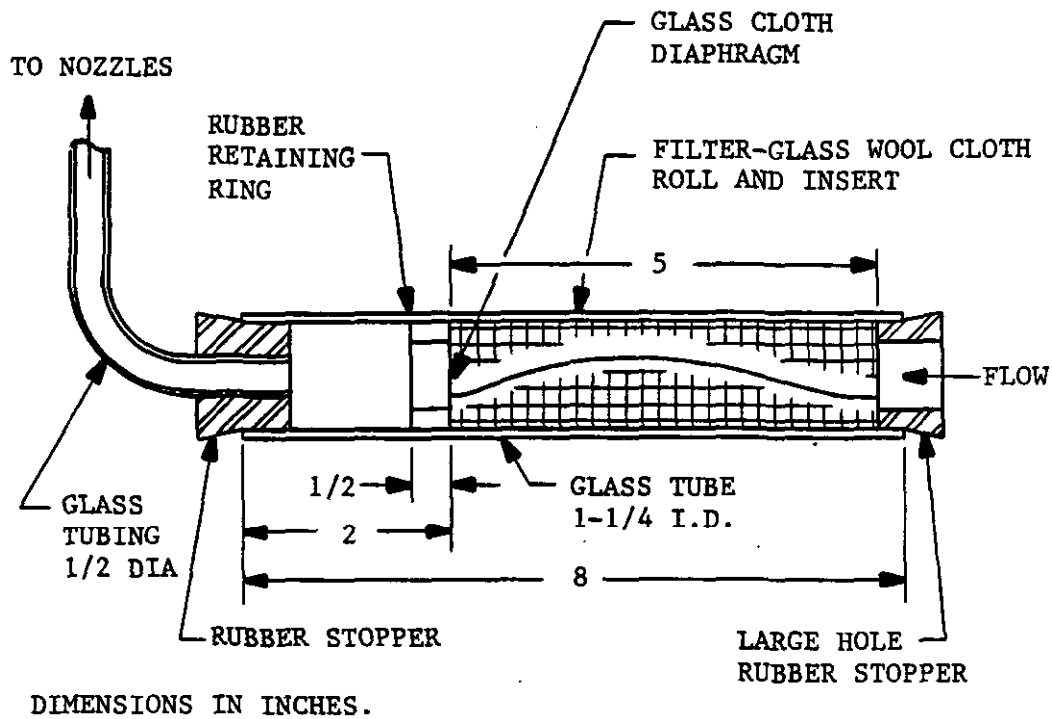
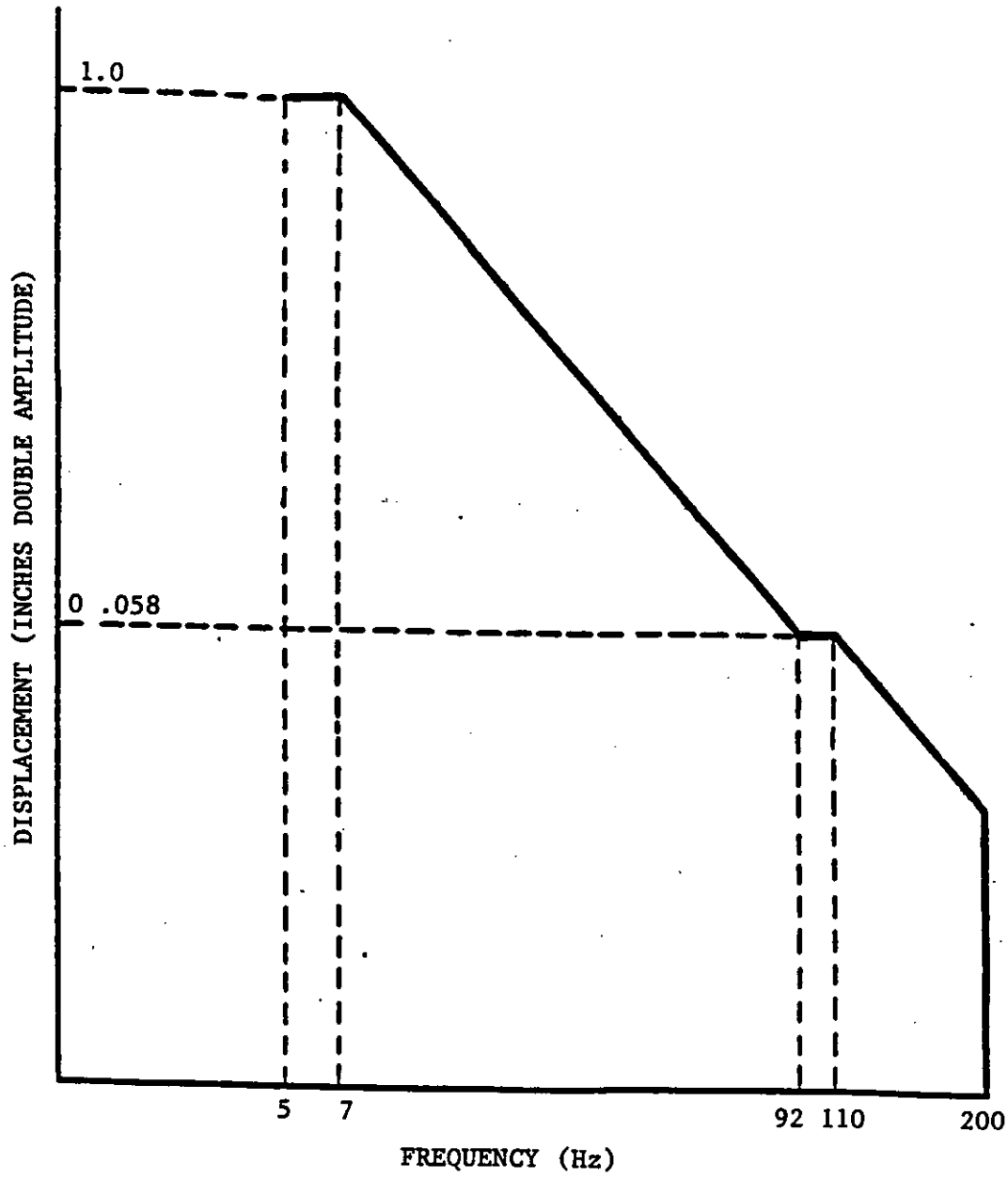


FIGURE 2. Salt solution filter.

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ACCELERATION LEVELS: ±g (PEAK)

FIGURE 3. Vibrational test curve.

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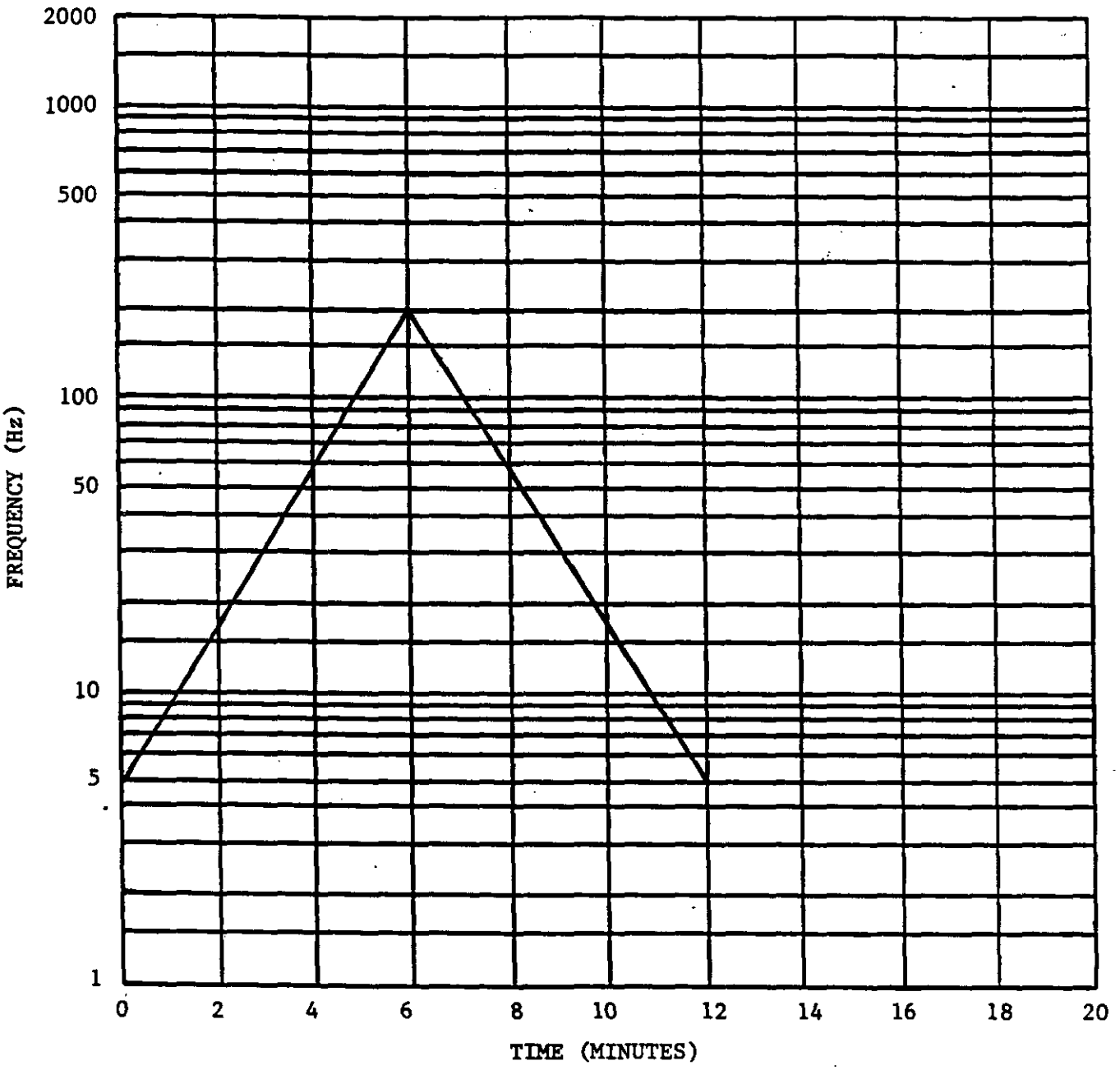


FIGURE 4. Logarithmic sweep.

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Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 4910-0681)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-62414(AT); Test Equipment (Simplified) Set Communicator (Set Com)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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